EXPERIMENT - 10

Application Performance Optimization

Library Book Issue Management

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Subject	Low-Code Application Development using Zoho Creator
Application	Library Book Issue Management
Platform	Zoho Creator
Record Date	16 October 2026

Aim

To apply query tuning and bounded subform input to the library project, and evaluate indexing considerations while preserving the required thirteen-field request form.

Experiment overview

Comparison of an application-side loop with a filtered aggregate, an indexing review, and a tested subform exercise. Query tests use the five existing Development requests.

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1. Problem Statement

A queue count does not need every request to be loaded and inspected in Deluge. Fetching all rows becomes unnecessary work as the library grows. A staff screen also needs only a short, ordered list of recent requests.

2. Tools and Data

Item	Value
Application / environment	Library Book Issue Management / Development
Source form	Book_Issue_Request_Form
Test function	Library_Query_Optimization_Test()
Dataset	5 requests: Pending 2, Under Review 2, Available 1
Test method	Read-only comparison with matching outputs; no latency benchmark

3. Implementation Procedure

1. Opened Workflow > Functions and created a no-argument void Deluge function.

2. Counted Pending requests by iterating over all request records as the baseline.

3. Replaced that operation with a criteria-based count aggregate.

4. Fetched only Pending request IDs, sorted by Added_Time descending, with range 0 to 2.

5. Executed the function and compared counts, visited rows and returned rows.

The range is inclusive, so 0 to 2 allows at most three records. Sorting makes the selected recent records predictable. No record is inserted, updated or deleted.

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4. Deluge Implementation

```
void Library_Query_Optimization_Test()
{
  // Read-only comparison; Development sample data only.
  baseline = 0;
  visited = 0;
  for each req in Book_Issue_Request_Form[ID != 0]
  {
    visited = visited + 1;
    if(req.Request_Status == "Pending")
    {
      baseline = baseline + 1;
    }
  }
  optimized = Book_Issue_Request_Form[Request_Status == "Pending"].count();
  recent = List();
  for each req in Book_Issue_Request_Form[Request_Status == "Pending"] sort by Added_Time desc
    range from 0 to 2
  {
    recent.add(req.ID);
  }
  result = Map();
  result.put("baseline_rows_iterated",visited);
  result.put("baseline_pending",baseline);
  result.put("optimized_pending",optimized);
  result.put("counts_match",baseline == optimized);
  result.put("bounded_rows_returned",recent.size());
  result.put("bounded_limit",3);
  result.put("writes_performed",0);
  info result;
}
```

5. Test Results

Check	Observed result
Baseline rows iterated	5
Baseline / optimized Pending count	2 / 2
Counts match	true
Bounded rows returned / limit	2 / 3
Writes performed	0

This verifies equivalent results and bounded retrieval. It does not measure milliseconds, database execution plans or large-scale throughput. A backend may still scan records even when Deluge does not iterate over them.

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6. Indexing Review

Creator manages database indexing behind the platform. Zoho documentation describes indexing as a backend process handled through support. No manual database index was created in this experiment.

Candidate	Decision for this library app
Record ID	Use the system record identifier for exact-record actions.
Request_Status	Use equality criteria for queue counts and reports. Request support review if a measured production workload becomes slow.
Preferred_Issue_Date	Keep the typed Date field for collection-date criteria.
Register_Roll_Number	Do not impose uniqueness: the same member can legitimately make multiple requests.

7. Bounded Subform Implementation

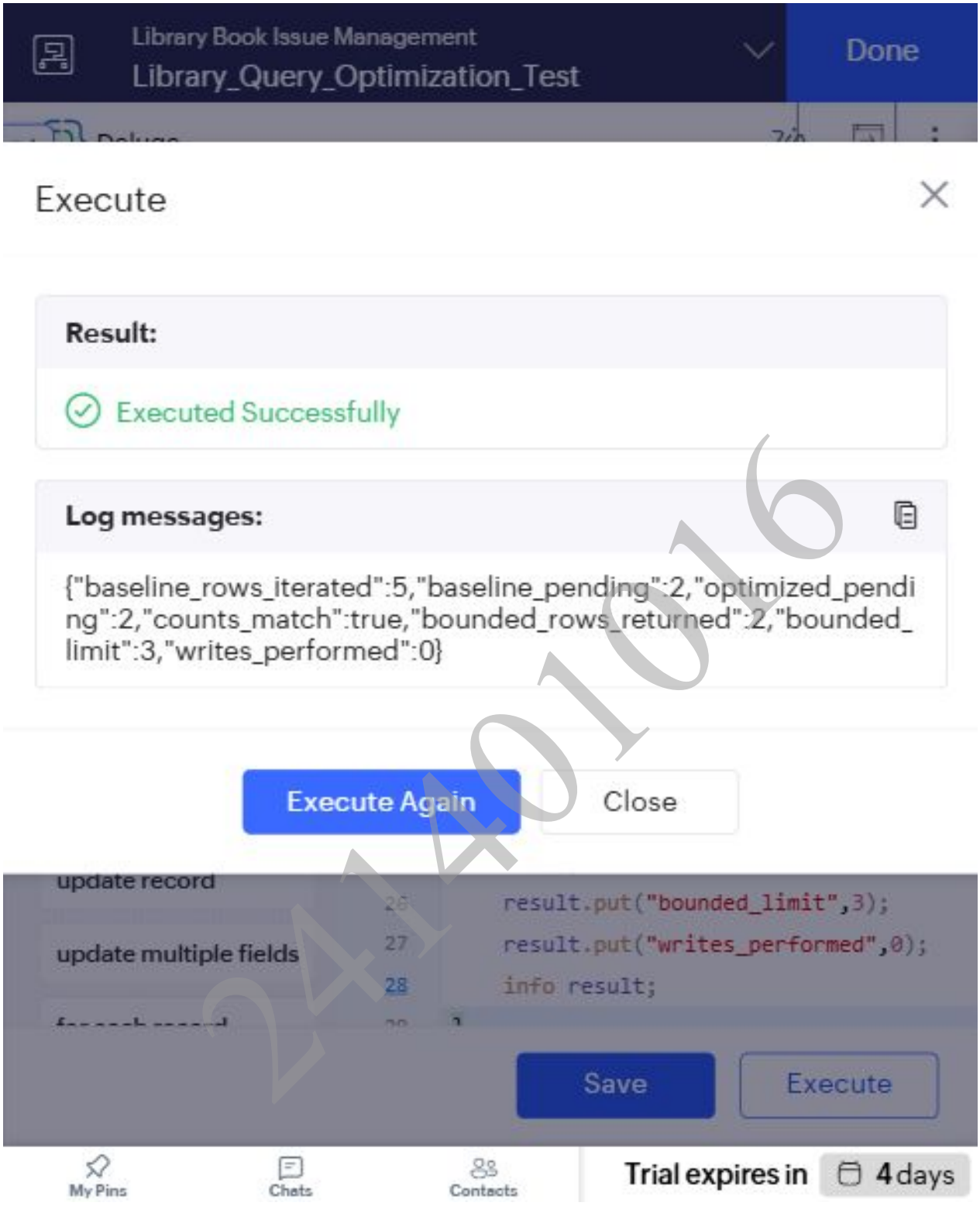
Created Library_Bulk_Request_Lab as a separate Development exercise. Its Lab_Reference is mandatory and unique. The mandatory Book_Lines subform contains mandatory Book_Title (Single Line) and Quantity (Number, default 1) fields. The maximum is three child rows.

Test	Actual result
LAB-SUB-001	One book line saved during initial testing.
LAB-SUB-002	Three books saved: Database System Concepts (1), The Pragmatic Programmer (2), Clean Code (1).
Attempt a fourth row	Creator rejected it: maximum number of rows reached.
Persistence	Both parent records and their book titles appeared in Library_Bulk_Request_Lab_Report.

The first attempt retained a one-row default. The limit was corrected, saved and retested at three rows. The exported schema confirms maxrow = 3. This exercise demonstrates bounded child input; it is not a complete inventory or lending subsystem. Quantity range validation and production access assignments would be separate additions.

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Implementation Evidence



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Figure 10.1. Development function execution: five baseline iterations, matching Pending counts of two, and a bounded result of two IDs. Captured 16 October 2026.

Implementation Evidence

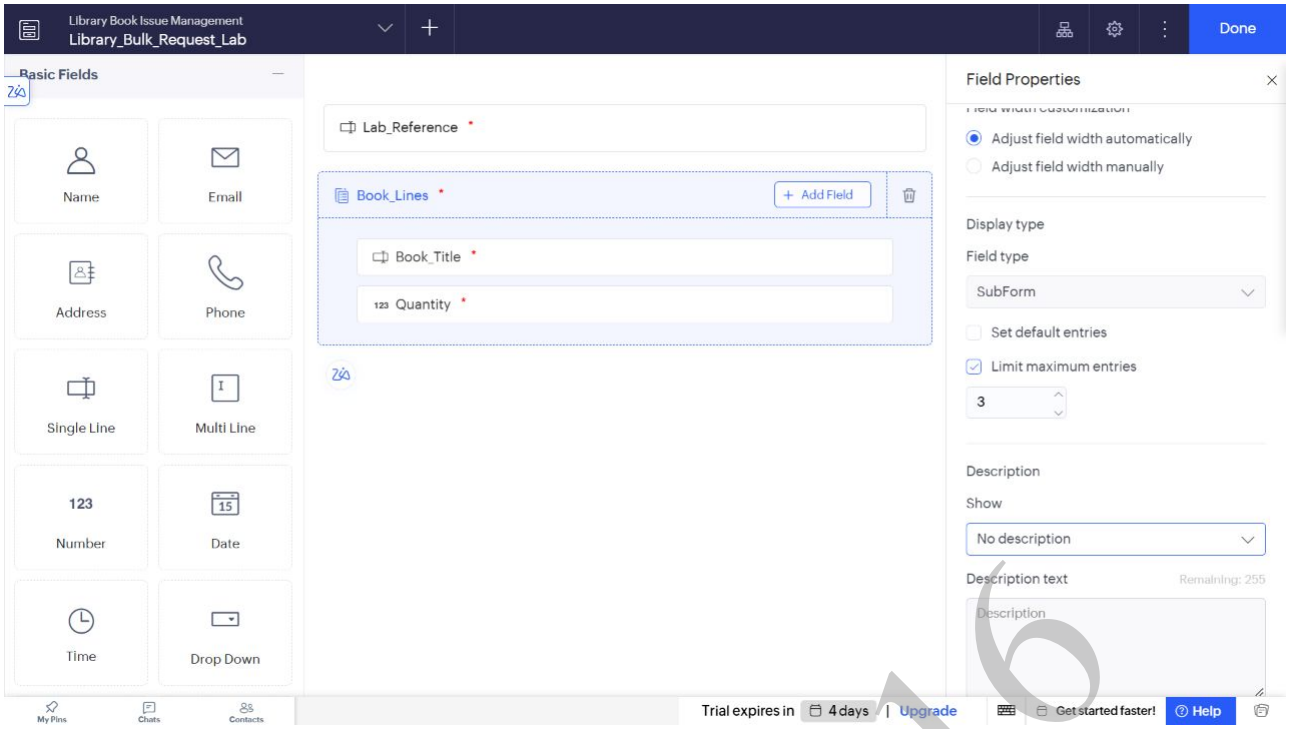


Figure 10.2. Development subform with mandatory child fields and a three-entry cap.

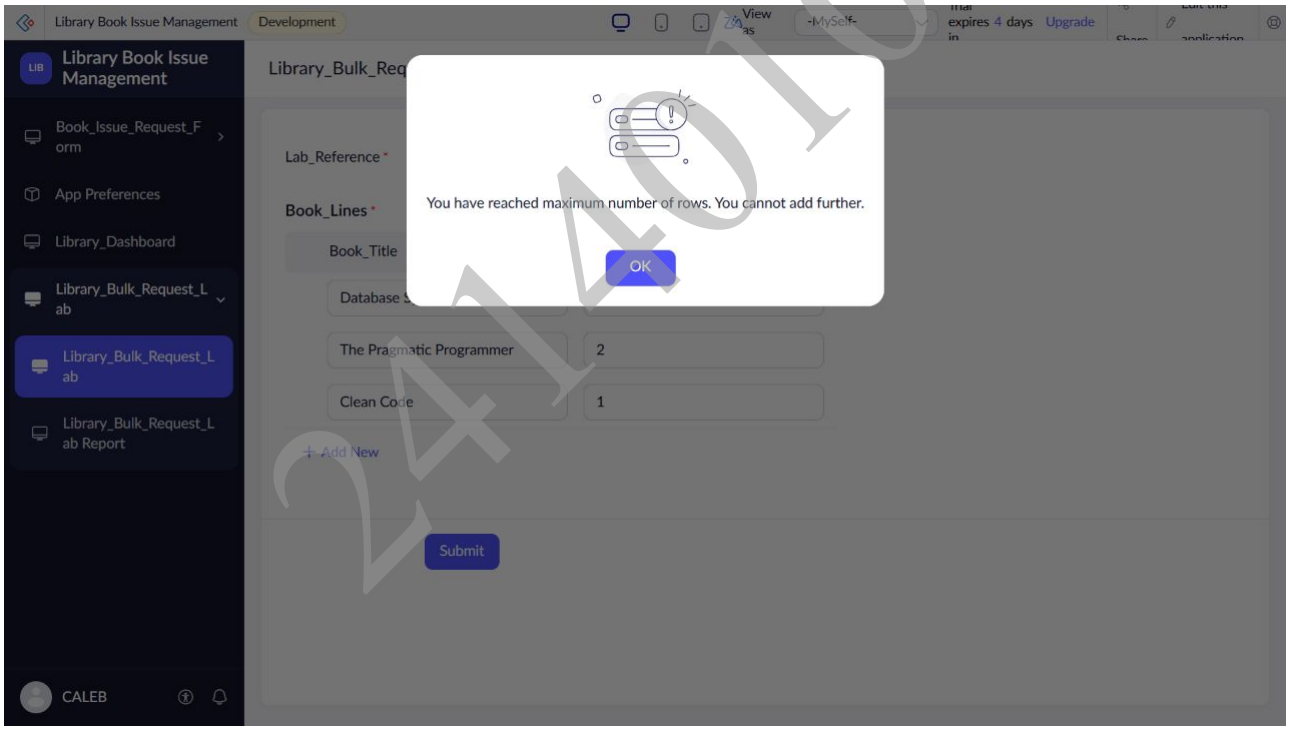


Figure 10.3. Actual negative test: a fourth child row is rejected after three titles have been entered.

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Implementation Evidence

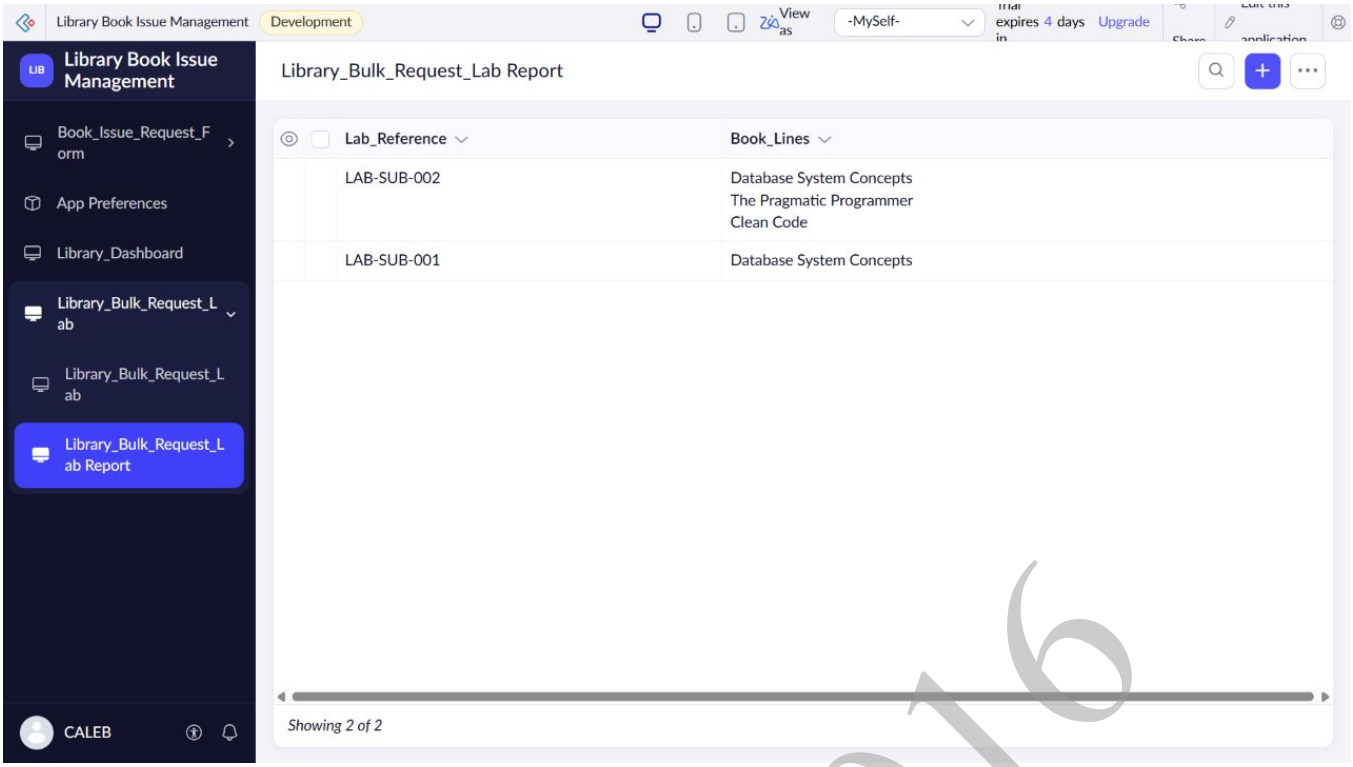


Figure 10.4. Saved parent records and child book titles in the automatically created lab report.

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8. Result

Query tuning and the bounded subform exercise were implemented and tested successfully. The criteria count matched the baseline, and the three-row cap was enforced. Indexing was reviewed as a managed backend capability; no manual database index was created.

9. Learning Outcomes

1. Use criteria-based aggregate functions for counts and totals.

2. Limit and sort records needed by a screen.

3. Separate correctness evidence from performance benchmarking.

4. Choose subforms only when the business relationship requires child records.

10. Further Performance Test

For a production-scale assessment, use approved synthetic data, compare median and slow-percentile response times under the same conditions, and inspect the slowest workflows. Record dataset size, concurrency, environment and API calls. Do not infer large-scale performance from five records.

References

1. Zoho Creator platform performance

2. Deluge: fetching and limiting records

Official sources checked on 16 October 2026. App observations and test results are identified separately.

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